

Promoting Two-sided Fairness with Adaptive Weights for Providers and Customers in Recommendation

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ABSTRACT

At present, most recommender systems involve two stakeholders, providers and customers. Apart from maximizing the recommendation accuracy, the fairness issue for both sides should also be considered. Most of previous studies try to improve two-sided fairness with post-processing algorithms or fairness-aware loss constraints, which are highly dependent on the heuristic adjustments without respect to the optimization goal of accuracy. In contrast, we propose a novel training framework, *adaptive weighting towards two-sided fairness-aware recommendation* (named *Ada2Fair*), which lies in the extension of the accuracy-focused objective to a controllable preference learning loss over the interaction data. Specifically, we adjust the optimization scale of an interaction sample with an adaptive weight generator, and estimate the two-sided fairness-aware weights within model training. During the training process, the recommender is trained with two-sided fairness-aware weights to boost the utility of niche providers and inactive customers in a unified way. Extensive experiments on three public datasets verify the effectiveness of *Ada2Fair*, which can achieve Pareto efficiency in two-sided fairness-aware recommendation.

CCS CONCEPTS

• Information systems → Recommender systems.

KEYWORDS

Recommender Systems, Fairness-aware Recommendation

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1 INTRODUCTION

In various online platforms, recommender systems play an important role in alleviating the information explosion [6, 17, 21], which can improve service experience and customer retention, thus increasing the expected profit of providers and promoting the long-term development of the platform. However, recommender systems also have negative aspects such as exacerbating the data bias [19] and amplifying the Matthew effect [1], so that dominant users and items will take the leading position and threaten the survival of disadvantaged groups and niche providers [18, 23], leading to a winner-take-all trend [3]. These side effects in recommendation are referred to as *fairness issues* [5, 13], in which unpopular providers obtain lower exposure and inactive customers get worse service.

In recommender systems, fairness has been studied in two major aspects, namely *provider-side fairness* and *customer-side fairness* [20]. For provider-side fairness, it refers to allocating relatively fair exposure to different providers [22], so that items from different providers can get reasonable exposure on the recommendation list. For customer-side fairness, it refers to reducing the performance discrepancy of different user groups or user individuals, so that all customers on a platform can get relatively satisfactory recommendation results [4, 19]. Since providers and customers are the main stakeholders in most recommender systems, we consider studying the *two-sided fairness*. By ameliorating the unfair situation from the two aspects, fairness-aware recommendation will promote sales for more providers and offer high-quality service to more customers.

Recently, a growing number of studies have started to focus on the two-sided fairness in recommender systems [13, 15, 22], and propose either post-processing algorithms or multi-objective constraints. Post-processing approaches typically simplify the fairness objective to a fair allocation [15] or combinatorial optimization problem [13, 22], and greedy algorithms are adopted to re-rank the existing recommendation results. However, post-processing methods are highly dependent on the heuristic re-ranking strategy and results of the trained recommender. Besides, due to a loose connection with the original recommender, it is difficult to entirely remove the underlying unfairness from the model, making it also infeasible to understand how the fairness occurs for specific models [1, 18]. Another line of methods introduces fairness-aware constraints into loss optimization by means of multi-task learning [5]. However, the additional fairness constraint (e.g., regularization of group utilities) is not consistent with the optimization goal of recommendation, leading to an unstable performance of various objectives [9].

Faced with these disadvantages, we promote fairness *with a closer connection* to the recommender. To this end, we propose a novel training framework, *adaptive weighting towards two-sided fairness-aware recommendation* (named *Ada2Fair*). The core lies in the extension of the conventional recommendation objective to a *fairness-aware objective* with learnable weights over the interaction data, focusing more on the samples from the minorities. In order to promote the fairness, we can accordingly adjust the importance of an interaction sample with the adaptive weight generator. Further, we estimate the two-sided fairness-aware weights during the training phase,

and a two-sided fairness-aware weight adapter is proposed to inject the fairness signals into the weight generator. During the optimization, the recommender is trained with two-sided fairness-aware weights to boost the utility of niche providers and inactive customers. Meanwhile, the original optimization recommendation objective is reserved to retain the accuracy. Moreover, our model-agnostic method can adapt to any model framework and fairness metrics, and extensive experimental results with *Pareto efficiency* have verified the effectiveness of our proposed Ada2Fair.

The contributions of this paper can be summarized threefold:

- As far as we know, it is the first attempt to promote the two-sided fairness in recommender systems via adaptive weights.
- We design a model-agnostic weighting framework towards two-sided fairness-aware recommendation (named *Ada2Fair*), which can balance accuracy and fairness in a unified way.
- Our experimental results demonstrate that Ada2Fair can promote two-sided fairness with Pareto efficiency over baselines.

2 PROBLEM AND DEFINITION

Problem Formulation. In this work, we study the problem of top- k item recommendation considering two-sided fairness for both providers and customers. Specifically, we denote $\mathcal{U} = \{u\}$ and $\mathcal{I} = \{i\}$ as sets of users and items respectively, where $|\mathcal{U}| = m$ and $|\mathcal{I}| = n$, and the user-item interactions between $\mathcal{U}$ and $\mathcal{I}$ can be written as a matrix $R \in \{0, 1\}^{m \times n}$. Meanwhile, we denote the items that user u interacted with as $\mathcal{I}_u = \{i \in \mathcal{I}, R_{u,i} = 1\}$. For provider-side fairness, we additionally consider the provider set $\mathcal{P} = \{p\}$ (with size l). We denote p_i as the provider of item i and $\mathcal{I}_p$ as the set of items belonging to provider p .

Based on historical interactions, the recommendation task aims to predict the probability of the potential interaction between user u and item i (i.e., $\hat{R}_{ui} = f_\theta(u, i)$, f is the recommender parameterized by θ). Then, for a user u , items with top- k predicted scores are recommended as the results L_u . Further, fairness-aware recommendation [5, 15] aims to *improve the fairness* and meanwhile *retain the accuracy*, so we consider three kinds of metrics with respect to *accuracy*, *provider-side fairness* and *customer-side fairness*. For accuracy, we adopt the widely used ranking based metric for evaluation, i.e., NDCG@ k [7]. In what follows, we introduce the metrics for *provider-side fairness* and *customer-side fairness*.

Provider-side Fairness. Existing literature [15, 23] generally assumes that provider-side fairness is achieved when the total exposure of providers is evenly distributed over the recommendation lists to users. Following the definition in [22], the exposure of an

item depends on its position in the top- k list, and the exposure of a provider is the average of exposure scores of all items it owns:

$$e_p = \frac{1}{|\mathcal{I}_p|} \sum_{i \in \mathcal{I}_p} \sum_{u \in \mathcal{U}} \frac{\mathbf{1}_{i \in L_u}}{\log_2(r_{u,i} + 1)}, \quad (1)$$

where $\mathbf{1}_{i \in L_u}$ equals to 1 when i is in the recommendation list L_u , and $r_{u,i}$ is the position of i in L_u which starts at 1, and we incorporate $|\mathcal{I}_p|$ as the denominator for normalization. To measure the deviation degree of the exposure of providers, we utilize the ratio of the standard deviation σ to the mean μ , also known as coefficient of variation (CV), which can eliminate the influence of measurement scale for performance comparison of different baselines. In this way, the provider-side fairness can be measured by CV as follows:

$$\text{P-Fair} = c_v(\{e_{p_1}, e_{p_2}, \dots, e_{p_l}\}), \quad (2)$$

where $c_v(\cdot) = \frac{\sigma(\cdot)}{\mu(\cdot)}$ means the coefficient of variation over a set of exposure scores. The lower it is, the more fairness it indicates.

Customer-side Fairness. As for customer-side fairness, we concentrate on the individual fairness of customers so as to improve the welfare of least-served users. In previous work [15, 22], it is acknowledged that to promote the individual fairness of customers is to balance the utility of different users (i.e., overall performance is aligned to long-tail performance). Same as the accuracy metric, we use NDCG to evaluate the utility of each user, and employ the standard deviation σ as the measurement of the dispersion degree:

$$\text{C-Fair} = \sigma(\{\text{NDCG}_{u_1}, \text{NDCG}_{u_2}, \dots, \text{NDCG}_{u_m}\}), \quad (3)$$

where we do not choose CV like Eq. (2) because the average value of NCDG is already reflected in the accuracy metric.

Note that fairness is a complicate concept, and there might be alternative methods to define P-Fair and C-Fair scores. Also, the bias factors of items (e.g., popularity) and the sensitive attributes of users (e.g., gender) are widely studied fairness-aware topics in academia. This paper jointly considers group fairness on the provider side and individual fairness on the customer side, and the two-sided fairness from more perspectives is worth further research in the future.

3 METHODOLOGY

3.1 Overview of Our Approach

Our main idea is to adjust the optimization scale through fairness-aware reweighing, thus achieving the *controllable preference learning* towards two-sided fairness. Specifically, we extend the original accuracy-focused loss by incorporating *fairness-aware weights*:

$$\mathcal{L}_{rec} = \sum_{u \in \mathcal{U}} \sum_{i \in \mathcal{I}_u} w_{u,i}^\lambda \cdot l_{u,i}(f_\theta), \quad (4)$$

where $\mathcal{I}_u$ denotes items that user u interacted with, $l_{u,i}(f_\theta)$ is the optimization loss between user u and item i on the recommender f parameterized by θ , and $\lambda \geq 0$ is the hyper-parameter to control the fairness level of preference learning. As the core of our method, $w_{u,i}$ determines the optimization scale of each interaction sample during the training process, which can control trade-offs between accuracy and different fairness metrics, regardless of whether the metrics are differentiable. Such an approach is compatible with various recommendation models (e.g., LightGCN [6] and SGL [21]) and optimization loss (e.g., BPR loss and cross entropy loss).

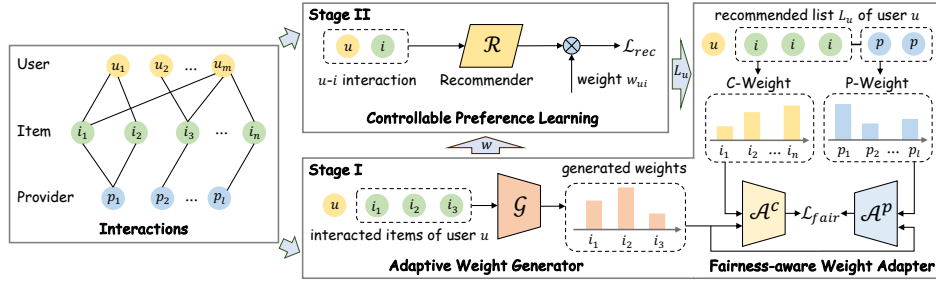


Figure 1: Overview of the proposed two-sided fairness-aware framework Ada2Fair.

3.2 Fairness-aware Weight Estimation

In order to promote the provider-side fairness, following Eq. (1), we propose to calculate the provisional exposure score e_p of each provider p based on the current *training state* (the model checkpoint of current iteration). As defined in Eq. (2), to promote the provider-side fairness is to balance the exposure scores among all providers. Based on e_p , we expect to increase the exposure score of under-exposed providers and decrease the exposure score of over-exposed providers. Therefore, we utilize the reciprocal of e_p as the provider fairness-aware weights:

$$\text{P-Weight} = \tilde{w}_{u,i} = \tilde{w}_{p_i} = \text{Norm} \left(\frac{1}{e_{p_i}} \right), \quad (5)$$

where $\text{Norm}(\cdot)$ means the linear normalization function for mapping weights between 0 and 1 to avoid exception values, and P-Weight denotes the provider fairness-aware weights.

As for the customer-side fairness, we expect to lower the performance discrepancy among different users. Based on the fact that the recommendation performance of one user is better when interacted items of the user are predicted to be ranked within the top positions with a higher probability [16, 19]. Therefore, we use the discounted cumulative gain [7] in the current rankings to estimate the user utility during the training process as $\tilde{e}_u$, and similarly employ the reciprocal of $\tilde{e}_u$ as the customer-side fairness-aware weights:

$$\tilde{e}_u = \frac{1}{|L_u|} \sum_{i \in L_u} \frac{1_{i \in L_u}}{\log_2(r_{u,i} + 1)}, \text{C-Weight} = \bar{w}_u = \text{Norm} \left(\frac{1}{\tilde{e}_u} \right), \quad (6)$$

where C-Weight denotes the customer-side fairness-aware weights.

3.3 Adaptive Weight Generator

Given the provider-side fairness-aware P-Weight ($\tilde{w}_{p_i}$) in Eq. (5) and the customer-side fairness-aware C-Weight ($\bar{w}_u$) in Eq. (6), the most straightforward way to inject two-sided fairness is to set P-Weight and C-Weight as $w_{u,i}$ in Eq. (4). However, such a way cannot be jointly optimized with the original recommendation objective. In our approach, we employ them as fairness guidance to learn the adaptive weight generator that can be optimized. Specifically, we take the vector $\mathbf{x}_u \in \mathbb{R}^n$ as input, *i.e.*, the u -th row of the interaction matrix $\mathbf{R}$. The generated weight $\mathbf{w}_u$ of user u is calculated as:

$$\mathbf{h}_u = \text{Tanh}(\mathbf{W}_1 \mathbf{x}_u + \mathbf{b}_1), \quad \mathbf{w}_u = \sigma(\mathbf{W}_2 \mathbf{h}_u + \mathbf{b}_2) = g_\phi(\mathbf{x}_u), \quad (7)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_1 \times n}$, $\mathbf{W}_2 \in \mathbb{R}^{n \times d_1}$ are trainable weight matrices. d_1 is the hidden size, $\mathbf{b}_1$ and $\mathbf{b}_2$ are bias terms, and Tanh and σ are the

non-linear activation functions. Since the design of neural networks is not our focus, we choose lightweight structures for efficiency. For simplicity, $\mathcal{G}$ is the weight generator parameterized by ϕ , and $\mathbf{w}_u \in \mathbb{R}^n$ is the generated weight vector over the item set for u .

3.4 Two-sided Fairness-aware Adapter

To consider fairness for both providers and customers, we utilize the simple one-layer neural network to reduce the dimension size of $\mathbf{w}_u \in \mathbb{R}^n$ from n to d . By mapping vectors to a tailored hidden space (each for a kind of fairness), adapters can capture the potential similarity between the generated and estimated weights as:

$$h_\chi(\tilde{\mathbf{w}}_u) = \text{Tanh}(\mathbf{W}_3 \tilde{\mathbf{w}}_u + \mathbf{b}_3), \quad h_\psi(\bar{\mathbf{w}}_u) = \text{Tanh}(\mathbf{W}_4 \bar{\mathbf{w}}_u + \mathbf{b}_4), \quad (8)$$

where $\tilde{\mathbf{w}}_u$ and $\bar{\mathbf{w}}_u$ are fairness-aware weights in Eq (5) and Eq (6), respectively. $\mathcal{A}^p$ (parameterized by χ) and $\mathcal{A}^c$ (parameterized by ψ) are the fairness-aware adapters for providers and customers, $\mathbf{W}_3 \in \mathbb{R}^{n \times d_2}$, $\mathbf{W}_4 \in \mathbb{R}^{n \times d_3}$ are trainable matrices for transformation.

Within the target hidden space, the weight adapters are used to minimize the gap between generated and estimated weights. For simplicity, we utilize the L2 loss to optimize $\mathcal{G}$ and the adapters:

$$\mathcal{L}_{fair} = (1 - \alpha) \|(h(g(\mathbf{x}_u)), h(\tilde{\mathbf{w}}_u))\|_2 + \alpha \|(h(g(\mathbf{x}_u)), h(\bar{\mathbf{w}}_u))\|_2, \quad (9)$$

where α is the coefficient to control the balance between the two-sided fairness. Through the joint training of $\mathcal{L}_{rec}$ and $\mathcal{L}_{fair}$, the generator $\mathcal{G}$ obtains fairness signals for the recommender to improve the controllable preference learning in Eq. (4), while the recommender $\mathcal{R}$ provides the current ranking list for $\mathcal{G}$ to guide the integration of two-sided fairness-aware reweighing in Eq. (9).

3.5 Comparison with Existing Work

To highlight the novelty of Ada2Fair, we make a brief comparison with existing work in Table 1. For popularity-biased approaches, PNS [1] and IPS [18] improve the provider-side fairness by negative sampling and inverse propensity scores. However, the applied static strategy has no adaptive ability during the training process. For two-sided fairness-aware methods, in-processing approaches such as Multi-FR [5] take fairness-aware recommendation as multi-objective optimization problems. However, simultaneously constraining each kind of fairness often involves conflicts. Besides, post-processing methods such as FairRec [15], TFRM [22] and CPFair [13] are proposed to adjust results towards fairness with a trained recommender, while the two-stage processing fails to improve fairness by the model itself. For time complexity, the time

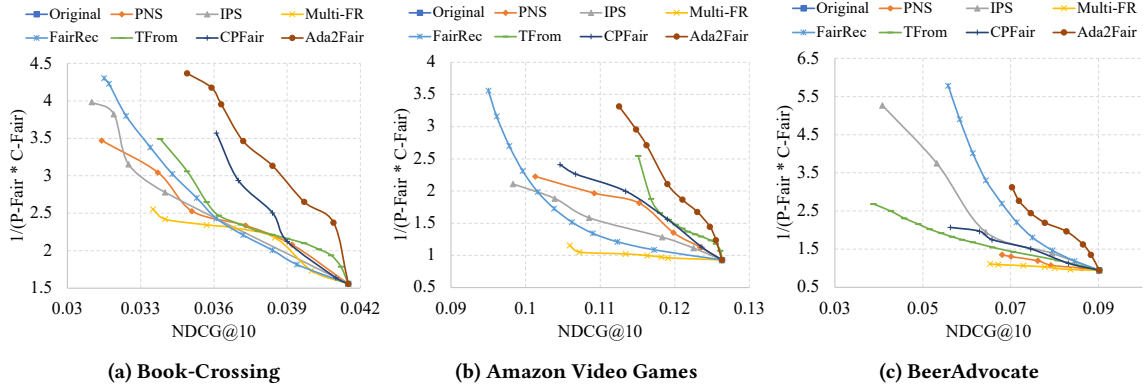


Figure 2: Trade-off curves between two-sided fairness (reciprocal of P-Fair and C-Fair) and recommendation accuracy (NDCG) with the BPRMF model. The closer the curve is to the upper right corner, the better the overall performance.

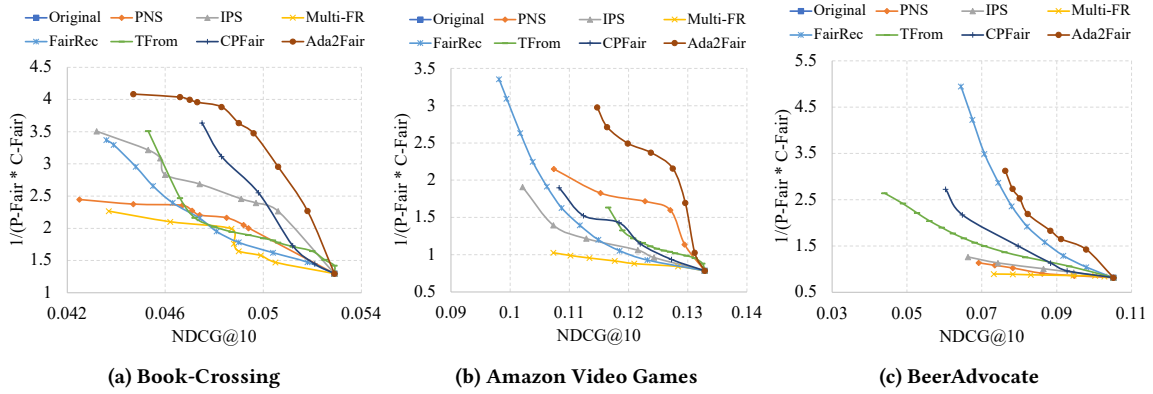


Figure 3: Trade-off curves between two-sided fairness and recommendation accuracy with the LightGCN model.

Table 1: Comparison of different methods. “PF” and “CF” denote provider- and customer-side fairness. “Adaptive” indicates that the model adaptively adjusts fairness, and “Unified” means the two-sided fairness is promoted in a unified way.

Method	PF	CF	Adaptive	Unified	Complexity
Original	✗	✗	N.A.	N.A.	$O(T)$
PNS [1]	✓	✗	✗	N.A.	$O(n + T)$
IPS [18]	✓	✗	✗	N.A.	$O(n + T)$
Multi-FR [5]	✓	✓	✗	✗	$O(mn + T)$
FairRec [15]	✓	✓	✗	✓	$O(mn + T)$
TFROM [22]	✓	✓	✗	✓	$O(nm^2 \log(m) + T)$
CPFair [13]	✓	✓	✗	✓	$O(mn + T)$
Ada2Fair (ours)	✓	✓	✓	✓	$O(mn + T)$

cost of weight generation depends on the number of users and the dimension of network, *i.e.*, $O(dmn)$. By setting $k, d \ll n$ and $l \leq n$, the complexity of stage I is $O(mn)$. As for the controllable preference learning in stage II, our weighed paradigm in Eq. (4) has no additional time cost compared to the conventional training strategy, *i.e.*, $O(T)$. Therefore, the time complexity of Ada2Fair, *i.e.*, $O(mn + T)$ is comparable to two-sided fairness-aware methods.

4 EXPERIMENT

4.1 Experimental Settings

Dataset Descriptions. We select three public datasets, *i.e.*, Book-Crossing [27], Amazon Video Games [14], and BeerAdvocate [12] to conduct experiments considering typical domains (books, games and beverages) and different scales. To evaluate the two-sided fairness, publishers in book domain, companies in game domain and brewers in beer domain are extracted from attribute information as the provider of items, which is consistent with the actuality. For datasets, we filter users and items with less than five interactions.

Evaluation Metrics. As introduced in Section 2, we choose the widely-used $NDCG@k$ as the metric ($k = 20$ as applied in [6, 21]) to evaluate the accuracy of recommendation, and P-Fair and C-Fair are utilized to measure the two-sided fairness. We compute all metrics based on all-ranking protocol [8] and the results are averaged over all users and providers, respectively. To evaluate overall performance of various metrics, a common solution is to define the harmonic average of accuracy and other metrics (*e.g.*, diversity) as F-score [2, 11]. Similarly, we propose a new AF score indicator based on the trade-off between bilateral fairness and accuracy. To

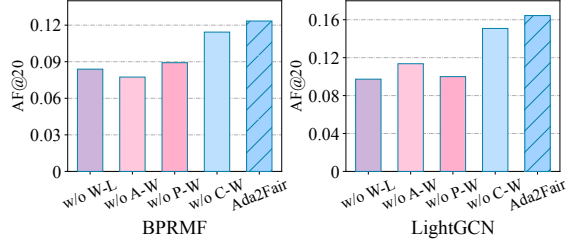


Figure 4: Ablation study of Ada2Fair on the Book-Crossing.

facilitate the comparison of baselines, we define an AF-score to measure the comprehensive performance within three metrics [2, 11]: $AF@k = \frac{NDCG@k}{P-Fair@k \times C-Fair@k}$. The higher the AF, the better the overall performance of accuracy and two-sided fairness.

Baselines. We compare Ada2Fair with baselines as mentioned in Section 3.5: **Original** is the default setting for training regardless of fairness. **PNS** [1] refers to Popularity-biased Negative Sampling, which prefers to select popular items as negative samples. **IPS** [18] is the Inverse-Propensity-Scoring (IPS) estimator, which assigns static weights according to the reciprocal of item popularity to each training sample. **Multi-FR** [5] is a multi-objective optimization framework for multi-stakeholder fairness-aware recommendation. **FairRec** [15], **TFROM** [22] and **CPFair** [13] are three post-processing algorithms towards two-sided fairness.

Implementation Details. We implement all models based on the open-source benchmark library RECBOL [24–26]. Since our method is model-agnostic, we select two representative methods BPRMF [17] and LightGCN [6] as the base recommender. As for general parameters of all methods, we tune the learning rate in $[1e-4, 5e-4, 1e-3, 5e-3, 1e-2]$ and the number of graph convolutional layers in $[1, 2, 3]$. For Ada2Fair, we tune α ranging in $[0, 1]$ to strike a balance between the two-sided fairness. For baselines, we carefully search the hyper-parameters following the settings of original authors. We provide our code at <https://github.com/RUCAIBox/Ada2Fair>.

4.2 Overall Performance

Pareto Efficiency. In our experiment, three aspects are involved: accuracy, provider-side fairness and customer-side fairness. And under different parameter settings, three metrics also fluctuate. In other words, fairness-aware recommendation is a trade-off between accuracy and fairness, and the degree of trade-off can be adjusted by parameters [13, 15, 22]. Therefore, comparing a single metric has no practical meaning, and comparing the performance of three metrics only under a certain setting is also unscientific. For a fair comparison between baselines, we visualize the change curves between metrics and comprehensively compare different methods based on the concept of *Pareto efficiency* [10]. That is to say, we obtain the changes in fairness metrics under different degrees of accuracy, in order to comprehensively evaluate the performance comparison between methods. As shown in Fig. 2 and Fig. 3, we can see that our proposed Ada2Fair achieves Pareto optimality over baselines considering both accuracy and fairness.

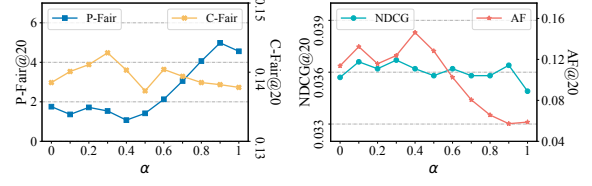


Figure 5: Performance comparison w.r.t. different α .

Ablation Study. To analyze how each of our proposed components affects the overall performance, we devise four variants of Ada2Fair including (1) w/o W-L that replaces controllable preference learning loss by conventional BPR loss, (2) w/o A-W replaces the adaptive weight generator by the average value of P-Weight and C-Weight, (3) w/o P-W without consideration of provider-side fairness and (4) w/o C-W without customer-side fairness. In Fig. 4, we can see all components of Ada2Fair contribute to the final results.

Impact of Coefficient α . In our proposed $\mathcal{L}_{fair}$ loss defined in Eq. (9), the coefficient α controls the importance weights between provider-side and customer-side fairness. In Fig. 5, we can observe that P-Fair has an upward trend while C-Fair has a downward trend with the increase of α , which shows that α has the ability to adjust the balance between provider-side and customer-side fairness. By adjusting α , we can find parameters that perform well in terms of both accuracy and two-sided fairness with a higher AF value.

5 CONCLUSION

In this work, we propose an adaptive weighting framework towards two-sided fairness-aware recommendation (named *Ada2Fair*) consisting of two key modules: controllable preference learning and fairness-aware weight generation. Specifically, we extend the conventional recommendation objective to a *fairness-aware objective* with adaptive weights. Considering the fairness-aware weights, we utilize an adaptive weight generator to adjust the importance of an interaction sample, and two-sided fairness-aware weight adapters are employed to inject the fairness signals. Extensive experimental results show that Ada2Fair can achieve Pareto efficiency towards two-sided fairness. For future work, we will further consider applying the paradigm of adaptive weighting to more recommendation tasks (e.g., sequential and social recommendation) and other aspects of recommendation (e.g., user privacy and diversity).

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